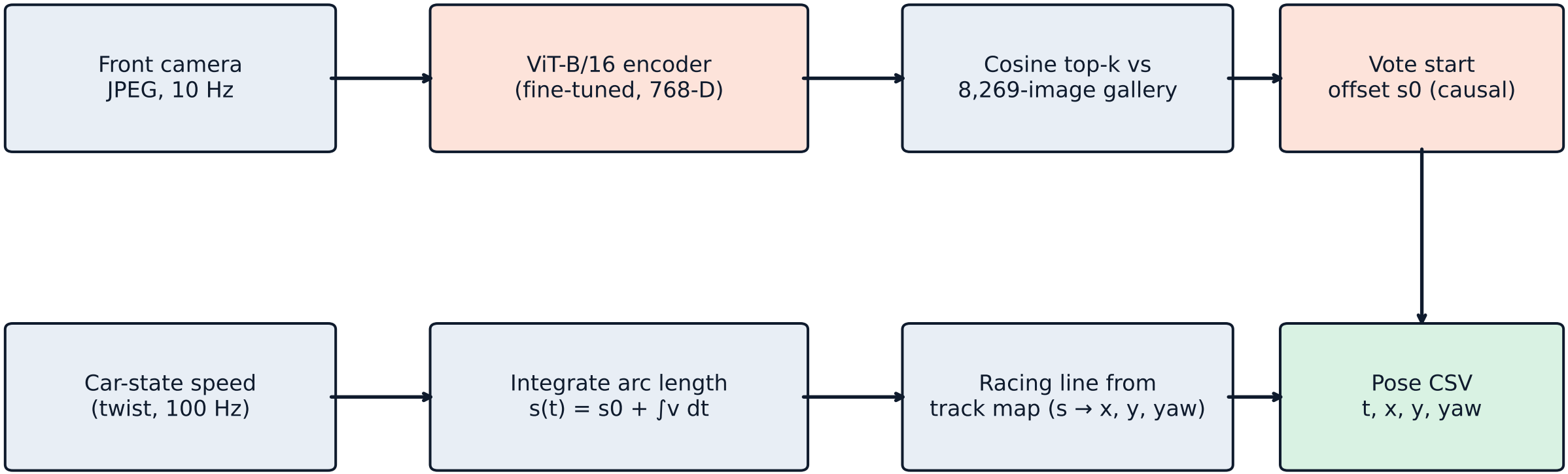
Camera-only localization at race pace

Learned place recognition + causal track-constrained odometry

- Input: front camera images, track map, car-state speed. No GNSS, LiDAR, IMU or ground truth at inference.
- ViT-B/16 place descriptors fine-tuned on training laps 1-5 (8,269 images).
- Causal: the pose at frame t uses only data up to t. 10 Hz output, 100 % coverage.
- Validation laps 6-7 (official scorer): position RMSE 1.97-2.64 m, lateral RMSE 1.26-1.35 m.
- Real-time on a laptop iGPU: 75 ms/frame encode + <1 ms tracking (Ryzen AI 7 350, Radeon 860M).

System



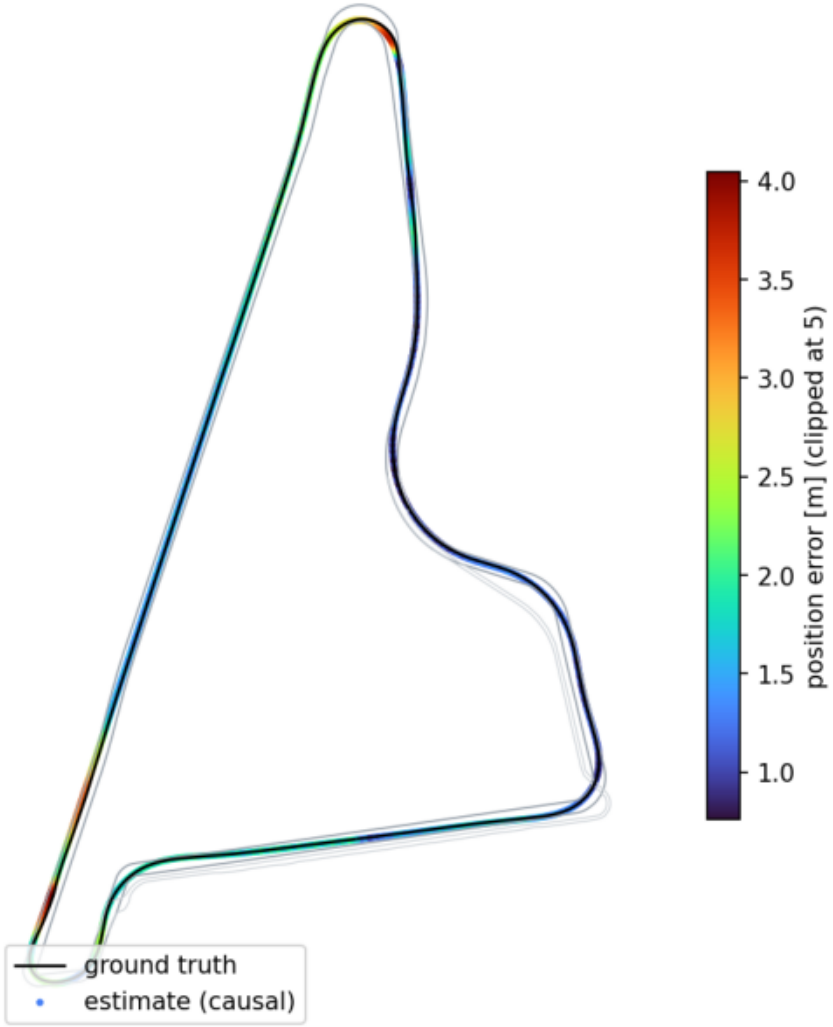
Training (offline): triplet fine-tuning of the last two ViT blocks on laps 1-5, cross-lap positives <=5 m/15°, negatives >=25 m.
Gallery = training-lap descriptors with their poses. Laps 6-7 held out for validation.

Validation results (official scorer, laps 6-7)

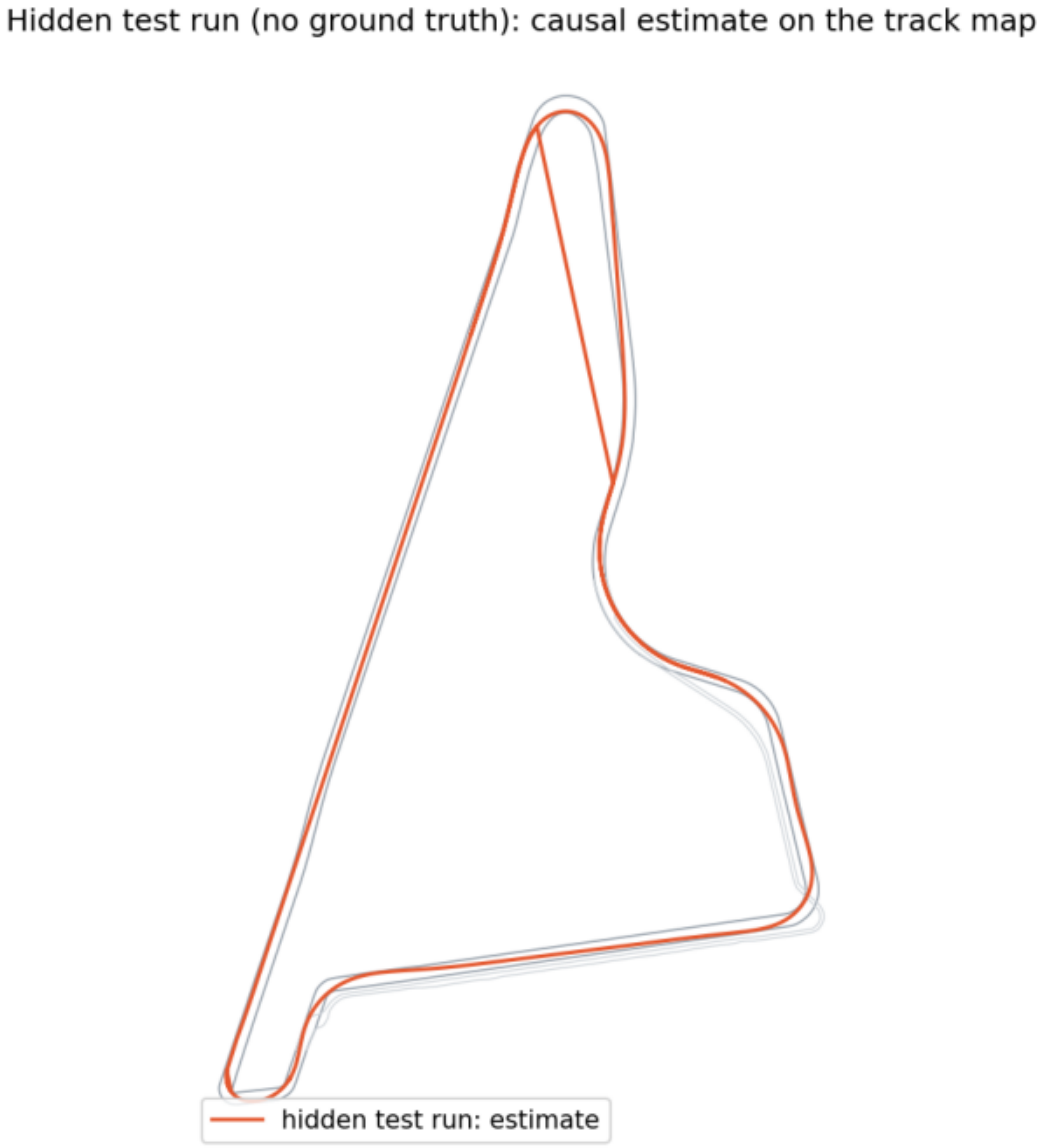
Method	Lap	Pos RMSE	Lateral	Yaw deg	<1 m
Track odometry (submitted)	6	2.64 m	1.35 m	2.72	19 %
	7	1.97 m	1.26 m	2.29	14 %
Multi-hypothesis tracker	6	1.79 m	0.63 m	0.79	69 %
	7	1.83 m	0.83 m	1.02	63 %
Retrieval filter, no motion	6	2.45 m	0.82 m	1.07	53 %
	7	2.91 m	0.84 m	1.31	56 %
Earlier team baseline	4-5	15.0 m	1.54 m	1.37	39 %

Coverage 100 % (1 s bins) on both laps. Speed for laps 6-7 is a GT-derived proxy (training run has no twist topic); the hidden test uses real car-state twist. Laps 6-7 were seen in earlier baselines: reserved validation, not pristine.

Validation lap 7: causal camera-only estimate vs ground truth



Hidden test run: daylight, up to 68 m/s



- 739 front-camera frames, 74 s, one full lap (2,964 m integrated speed).
- Training data is night only. Daylight retrieval alone jumps between track sections.
- Fix: retrieval only votes the start offset; speed integration carries continuity; hysteresis prevents flips.
- One correction jump in the first seconds while votes accumulate; then a continuous lap.
- No first pose was used. 739/739 frames emitted (coverage 100 %).

Runtime and engineering

- Laptop: AMD Ryzen AI 7 350, Radeon 860M iGPU (DirectML), 32 GB RAM, Windows 11.
- ViT-B/16 encode 75 ms/frame (batch 12) + tracker 0.6 ms/frame: about 13 Hz, faster than the 10 Hz camera.
- One run script: localizer/kiro_localize.py (encode + gallery) and localizer/track_odometry.py (causal pose).
- Fine-tuning pipeline with RAM/time guards, durable checkpoints and full held-out validation (backend/).
- Weights (343 MB) as a GitHub release asset; data stays local per partner licence.

Failure cases and lessons learned

- Night-to-day domain shift is the main failure: daylight descriptors match wrong but similar straights.
- Repetitive long straights cause longitudinal error (1.1-2.3 m RMSE dominates position error).
- Track constraint pins lateral error to the racing-line offset; true lateral deviation is not observed.
- Planned, not done: lighting augmentation, lateral offset from lane borders, multi-camera descriptors.
- Lesson: a time budget must be checked before expensive steps; our last 1-hour training run spent its budget on baseline encoding and pair mining and produced 0 new updates, so the submitted model is step 7750.